

Comparing Free Reference Extraction Pipelines

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Abstract

Purpose: In this paper, we compare the performance of several popular pre-trained reference extraction and segmentation toolkits combined in different pipeline configurations on three different datasets. **Methods:** The extraction is end-to-end, i.e. the input is PDF documents, and the output is parsed reference objects. The evaluation is for reference strings and individual fields in the reference objects using alignment by identical fields and close-to-identical values. **Results:** Our results show that Grobid and AnyStyle perform best of all compared tools, although one may want to use them in combination. **Conclusion:** Our work is meant to serve as a reference for researchers interested in applying out-of-the-box reference extraction and -parsing tools, for example, as a preprocessing step to a more complex research question. Our detailed results on different datasets with results for individual parsed fields will allow them to focus on aspects that are particularly important to them.

Keywords: reference extraction, reference segmentation, toolchain, parsing, scholarly document processing

1 Introduction

Citations are one of the most prominent features of scientific literature. They implement the very core of the scientific process, i.e., the idea of standing on the shoulders of giants by crediting and referencing previous work. Although this may not have been entirely obvious from the start, scientific references are not only interesting to the reader of the publication that contains them. Any scientific publication is usually designed to maximize its utility for future work, often approximated as the number of references it receives from other publications. As the future references are not known at the time of the publication of the referenced work, it is not possible to record the publications that cite a certain work (i.e. the forward look on the *citing* works) within this work

– although this may be equally as relevant as the conventional backwards looking record on the *cited* works. In addition to the number or the exact list of citing documents given a specific work, the entire network of referenced documents or a particular strain may interest people ignorant of the content of the individual publications therein. Such researchers may be referred to as *bibliometricians* or *scientometricians*.

As the first of their kind, *Eugene Garfield* is generally credited with having founded the discipline of recording and studying scientific citation networks after being inspired by *Shepard's Citations*, a similar solution for American court cases [66]. Starting with his *Science Citation Index* [17] citations or references have been extracted from the publications they appear in and collected in external resources ever since. This was initially

conducted manually and is still done in this way, although the effort has probably been moved to the authors of publications as they are asked to provide a list of references when publishing their work. In addition, the vast growth in the volume of new scientific literature [36] makes it attractive to automatically extract this information when manually curated references are unavailable. This is supported by the fact that many citation indices continue to be proprietary, largely overlapping [52], and expensive to use [29]. Instead, automatic extraction offers the option to obtain a set of references given the publication in electronic text (like UTF, ASCII), structured form (like DOC, XML, HTML), or print medium (like PDF, PS).

Many tools have been proposed for automatic reference extraction from scientific publications. Some tools may only be able to process certain input types (well). In addition, many different ways of presenting scientific work in general and referenced work in particular exist. Venues have different page formatting and reference styles, including footnote referencing. Considerable high-level differences in conventions of different kinds may exist between different scientific domains. Some tools may be more suitable for some of them.

In this work, we offer decision support for choosing reference extraction tools. We conduct an empirical evaluation of different toolchains applied to PDF publications, mainly from the social and geosciences. These publications have been manually processed to extract and parse their references, which allows for reliable evaluation of the performance of the compared tools. We distinguish the task of extracting simply the reference string from its subsequent parsing into a reference object, i.e., a set of meta-information on the referenced publication. In contrast to other comparisons, our contributions are focused on:

1. ... observing toolchains, i.e. combinations of tools, not just individual ones
2. ... evaluating based on individual reference object fields
3. ... using an advanced mapping procedure to match extracted and gold fields

In the following, we discuss related work in Section 2 and the compared tools in Section 3, describe our experimental setup in Section 4 and the corresponding results in Section 5 before we conclude in Section 6.

2 Related Work

We briefly summarize the research on reference extraction and parsing. The tools that we have been able to identify as being presented in the literature are listed in Table 1 along with their name, the year of the publication, whether they extract or parse references, and how they do so.

2.1 Reference extraction

Reference extraction and parsing is a special case of information extraction. Classical information extraction (including all reference extraction methods listed in this paper) use traditional techniques like multivariate classifiers (e.g. *Support Vector Machines* (SVMs) [65] or sequence labelling (e.g. *Conditional Random Fields* (CRFs) [35], or *Hidden Markov Models* (HMMs) [18], or *Structured SVMs* [60]) to produce a sequence of labels for a given input text sequence. Likewise, a multipurpose generative model will produce a natural language text sequence for an input sequence. Given recent breakthroughs in generative AI using *Large Language Models* (LLMs), various groups have experimented with re-purposing LLMs for information extraction. [1, 3, 5, 37, 51, 62, 64, 70] In the most popular instance of such generative models, namely a chat bot, the input sequence is considered a question or instruction and the output a natural language answer. When using a chat bot like ChatGPT for information extraction, the input needs to contain both the instruction (e.g. ‘extract all references’) and the text to extract from (e.g. a part of a publications full-text) and the output needs to be turned into a machine-readable, standardized format. Given the unnecessary overhead of turning a formalized problem such as reference extraction into a natural language query and then again interpreting a natural language output, it is more efficient to consider which aspect the LLM is really contributing compared to traditional approaches, and isolating those, while still using the fast and predictable traditional methods for labelling. [25] For example, the LLM’s vast background knowledge may include aspects of reference matching, correction and completion out-of-the-box (cf. [14, 63]).¹

¹Yann LeCun’s post on LinkedIn may also be of interest: <https://tinyurl.com/w4yk5nmm> [accessed 3rd March 2024].

One way of combining an LLM’s natural language capacities and background knowledge with a dedicated labelling method is to (fine-)tune an auto-regressive model specifically to take text sequence as input and output only the desired formal output. [31] Another way is using an LLM’s ability to creatively produce varied correct natural language texts (with annotation) to produce training samples (e.g. texts with made-up annotated references or parsed references) in a much cheaper way than using human annotators. This allows to create very large and variable silver-standards that can in turn be used to train task-specific compact classical models. [19, 55, 61] At this point, we have not been able to identify scientific works that use LLM’s to obtain reference extraction or parsing tools. We have experimented with using Llama2 prompts to extract references, but we have noticed issues with hallucination as well the high cost of having to enter the entire full-text with chopped substrings as individual prompts. There are plenty of scientific works introducing reference extraction and parsing methods. We have collected all the works that we could find in Table 1. Some only extract reference strings, some only parse reference strings, and some do both. Some of these have become known under a specific name, such as *ParsCit*, *Grobid* or *Cermine*. For most of the scientifically described methods, tools or working code cannot be (readily) retrieved, and many have never been given a name under which they may be found. Many methods are very old. Some published tools are described in the literature, while others are not. Of the fewer methods available as tools, many are not maintained anymore. In the following section, previous comparisons of tools are discussed. Those studies evaluated some of the listed tools as poorly performing.

2.2 Tool comparison

We have identified seven publications concerning the comparison of reference extraction software. Lipinski et al. [38] compare the tools *Doccar’s PDF Inspector*, *Grobid*, *Mendeley Desktop*, *ParsCit*, *PDFMeat*, *PDFSSA4MET*, *SciPlore Xtract*, *SVMHeaderParse* and *Zotero* for the task of extracting bibliographic metadata. Although many of the tools also extract references, this is not the focus of their comparison. Instead, they

are looking at metadata features of the publication (not of its references), i.e. title, authors, abstract and publication year. Here Grobid stands out with the best performance, but Mendeley Desktop and SVMHeaderParse also show acceptable performance. Körner et al. [34] compare tools specifically for reference extraction, however, look only at the reference string extracted, not its parsed version. Their tool RefExt outperforms all other tools², but Grobid follows in second place and Cermine as well as ParsCit also offer acceptable results. Complementing this research, Tkaczyk et al. [57] compare tools for the next step, namely the parsing of reference strings into nested reference objects that contain the metadata (like title, publication year and author names) provided for these publications in the references themselves. They list *AnyStyle*, *Biblio*, *BibPro*, *Cermine*, *Citation*, *Citation-Parser*, *Free_cite*, *Grobid*, *Neural ParsCit*, *ParsCit*, *PDFSSA4MET*, *Reference Tagger* and *Science Parse* and evaluate all of them except BibPro, Free_cite and Neural ParsCit, which they could not get to run. The results show an outperforming Grobid, followed by Cermine and then ParsCit. Mediocre performance is reported for AnyStyle³, Reference Tagger and Science Parse. Biblio, Citation, Citation-parser and PDFSSA4MET show poor performance. In a similar study, two additional aspects were studied in the same year. (A) Tkaczyk et al. [59] present a recommender for citation parsers, which presents a ranking of parsers for a given reference string such that the likelihood of a better-performing parser being ranked before another is maximized. Picking the top-recommended parser for any given reference string is shown to improve parsing performance compared to the single best-performing parser (Grobid). Apparently, some parsers work better for specific references than others, and these types of references can be detected to some degree by the used features (reference string length, number of commas, number of parentheses, etc.). Notably, they do not study the option to run all parsers on all reference strings and then combine their results optimally. (B) Tkaczyk et al. [58] compare the performance gains achieved over

²We have not been able to reproduce this.

³In our experiments, AnyStyle was a strong parser, so, likely, there have since been considerable improvements.

the pre-trained models when retraining a reference parser on gold data closer to the test data. While Grobid is already very strong in its pre-trained version and only gains 3 percentage points, ParsCit and Cerminе gain considerably by retraining them. In particular, Cerminе reaches the same F1 as a retrained Grobid. Probably, their models are very similar (both are CRFs), and the data used to train them is the main difference between their pre-trained versions.

Indrawati et al. [30] also compare Grobid, Cerminе and ParsCit, specifically to the case of Indonesian Scientific Journals. The results suggest that Grobid and Cerminе perform best, closely followed by ParsCit. Recently, Cioffi and Peroni [7] have compared end-to-end tools that do both reference string extraction and subsequent reference string parsing. They compare AnyStyle (preceded

Table 1: Reference extraction and parsing tools that are described in scientific literature or are available as tools. Issues: *availability*—no code found; *running*—problems running code; *installation*—problems installing; *quality*—poor results reported; *legacy*—officially abandoned; *proprietary*—not free-to-use.

publication	name	year	type	technique	code	issues
Constantin et al. [9]	PDFX	2013	extract	style-analysis	?	availability
Cuong et al. [13]		2015	extract	CRF	?	availability
Kern et al. [32]	TeamBeam	2012	extract	style-analysis	?	availability
Körner et al. [34]	RefExt	2017	extract	CRF	java	✓
Peng and McCallum [46]		2006	extract	CRF	?	availability
Councill et al. [12]	ParsCit	2008	extract+parse	regex+CRF	perl	running
Guo and Jin [23]		2011	extract+parse	KB	?	availability
Gupta et al. [24]		2009	extract+parse	regex+KB	?	availability
Lopez [39]	Grobid	2009	extract+parse	CRF	java	✓
Tkaczyk et al. [56]	Cerminе	2014	extract+parse	CRF	java	✓
Zou et al. [71]		2010	extract+parse	CRF+SVM	?	availability
Chen et al. [6]	BibPro	2010	parse	sequence alignment	java	installation
Cortez et al. [10]	FLUX-CIM	2007	parse	KB	?	availability
Cortez et al. [11]		2009	parse	KB	?	availability
Day et al. [16]		2007	parse	KB	?	availability
Hetzner [26]		2008	parse	HMM	python	availability
Hsieh et al. [28]		2014	parse	sequence alignment	?	availability
Kim et al. [33]	BILBO	2012	parse	CRF	?	availability
Namikoshi et al. [40]		2017	parse	CRF	?	availability
Ohta et al. [41]		2012	parse	CRF	?	availability
Ohta et al. [42]		2014	parse	CRF	?	availability
Ojokoh et al. [43]		2011	parse	HMM	?	availability
Staelin et al. [53]	Biblio	2007	extract	SVM+NN	perl	quality
Prasad et al. [48]	Neural ParsCit	2018	parse	NN	python	running
Suryawati and Widyantoro [54]		2017	parse	regex+rules+classif.	?	availability
Yin et al. [67]		2004	parse	bigram HMM	?	availability
Zhang et al. [68]		2011	parse	CRF	?	availability
Zhang et al. [69]		2011	parse	SVM	?	availability
Groza et al. [22]		2012	segment+parse	CRF	?	availability
pypi → citation-parser	citation-parser	2019	parse	?	python	running
AnyStyle.io	AnyStyle	2024	parse	CRF	ruby	✓
code4lib → FreeCite	FreeCite	2009	parse	?	ruby	installation
github → reftagger	RefTagger	2015	parse	?	python	legacy
github → pdfssa4met	pdfssa4met	2013	extract+parse	?	python	quality
github → science-parse	SPv1	2019	extract+parse	?	java	legacy
github → spv2	SPv2	2018	extract+parse	?	python	legacy
scholarcy.com	Scholarcy	2024	extract+parse	?	API	proprietary

by *pdftotext*), *Cermine*, *EXparser*, *Grobid*, *PDFSSA4MET*, *Scholarcy* and *Science Parse* and distinguish three task levels: (a) extract the set of reference strings, (b) determine the set of labels that the parser needs to assign to parts of a given reference string and (c) determine the set of part-label pairs for this reference string. We also use their gold standard in our evaluation. Similar to our experience and different from Tkaczyk et al. [57], AnyStyle performs well in their evaluation. The best full results are obtained by AnyStyle, Grobid and Cermine. AnyStyle (slightly) outperforms in each individual task. EXparser only works well in the second task. Interestingly, AnyStyle dominates the extraction task, although it cannot use visual layout information when simply using the text input from *pdftotext*. PDFSSA4MET continues to strongly underperform in this comparison. Science Parse shows mediocre performance compared to which the proprietary Scholarcy is better, but not as good as the top three ‘usual suspects’ AnyStyle, Grobid and Cermine.

Over all comparisons done over the years, the outstanding tools seem to be Grobid, Cermine, AnyStyle and (Neural) ParsCit. Soft weight-based or probabilistic models such as conditional random fields or neural networks outperform rule- or regex-based approaches, especially for reference string extraction.

2.3 Datasets and (re-)training

For training, tuning and testing end-to-end reference extraction and parsing, two types of datasets are required: (a) those that align input documents (e.g. in PDF format) with the reference strings that can be found in them; (b) those that align reference strings with parsed reference objects. The assumption is that the aligned outputs constitute the correct target for training, testing and tuning. The systems must, therefore, produce outputs of the same format, such that the deviation between the system output and the gold standard can be compared in terms of the system’s performance. The closer the system output is to the gold output, the better its performance. A system is considered better than another if it performs better on such datasets. Large datasets of the above kind benefit both the quality of the systems by offering more training examples and the significance of their comparison, so several efforts have been made to

create them. Colavizza and Romanello [8] have annotated references in fulltext documents about the history of Venice. Their annotation includes manually parsed reference objects and identifying the reference strings. They have used it to train CRF reference parsers and extractors. Grennan et al. [21] have created a large set of references in different reference styles by implementing a reference generator that takes publication metadata as input and outputs the respective reference strings for different reference styles. This does not seem to include any alignment between documents and the references mentioned therein and is hence of limited use to the extraction task.⁴ It is mainly advertised for training reference string parsers. In a related publication, Grennan and Beel [20] explore the use of this dataset for re-training Grobid and comparing the performance of the re-trained model against its out-of-the-box counterpart. Unsurprisingly, the results are that each model performs best on the data it was trained on and that for each dataset, the model that was trained on it performs best. For a third dataset, both models performed equally well. The training data used as an alternative to the original Grobid data was an equally large subset of the synthetic dataset mentioned above. Larger subsets did not consistently improve the results a lot. This should question, to some extent, the usefulness of very large datasets. Peroni and Shotton [47] have created and are maintaining *OpenCitations*, which is a large database that lists reference objects for many publications Daquino et al. [15]. It is our impression that the reference lists are not necessarily complete. Therefore, using this dataset to assess the precision of extraction tools should be difficult. The datasets used by us for evaluation are GEOcite by Birkeneder et al. [2], ExCite by Hosseini et al. [27] and CIOFFI by Cioffi and Peroni [7]. The latter is not freely available due to some PDFs not being open-access. However, it has been appended with a new open-source section by Pagnotta [44] for future use.

⁴It should be noted that a simple collection of reference strings or reference objects without the alignment to the original input document can also be used to train an extractor. This is because the collection can be used to learn a model of how references look like in contrast to normal text. Then, any input document can be scanned for substrings that match the learned patterns exceptionally well. However, this does not allow for tuning and evaluation.

3 Compared tools

We need to limit our selection as we are comparing not simply different tools but different combinations thereof. We do not compare tools or methods that are officially abandoned, that cannot be easily installed or run (e.g. requiring python2), that are behind a paywall, that do not have a machine interface or that have already been ruled out as poorly performing in previous comparisons. These issues are also listed in Table 1. Therefore, we only compare Grobid, AnyStyle, Cermin and ExParser. These have different input and output options, as listed in Table 2, which allows for different pipeline arrangements. We also include *pdftotext*, which returns the text from a given PDF for completeness. For example, AnyStyle can search for references in fulltext extracted by *pdftotext*. Generating all possible combinations, we end up with one pipeline for each path in Figure 1.

3.1 EXparser

EXparser [4], developed in the scope of the EXCITE project [27], is a tool based on machine learning. Its distinctive feature is that it aims to handle high-variance references. High-variance means references of a wide range of formats and locations, e.g., the tool can handle references in footnotes. EXparser relies on manually extracted textual and layout features for extraction and segmentation. Random Forest is used as a classifier for extraction, and Conditional Random Fields are used for segmentation. The final step in the process is checking the completeness of a reference, which is estimated by probability density functions on segmented reference strings.

3.2 Grobid

Grobid [39] is arguably the most popular tool for end-to-end reference extraction and parsing. It is

still maintained and can extract parsed references when given a PDF file⁵. The results consistently rank among the best when compared to other tools. References are only one type of information that it extracts. It also finds images, paragraphs, titles, authors and many other features, creating a completely structured representation of the publication. It can also be used to extract fulltext from a PDF, although this information is spread across many different fields. Finally, it also returns the reference strings in unparsed form.

3.3 Cermin

Cermin [56] is similar to Grobid. The difference seems to be slightly worse performance in its pre-trained form (c.f. Tkaczyk et al. [58]), slower speed and different input and output options. Cermin can also simply parse given reference strings and provides layout files for a given PDF.

3.4 AnyStyle

AnyStyle is, first and foremost, a parser for reference strings, but it can also discover them in surrounding fulltext. It cannot directly work with PDFs, which requires *pdftotext* or another fulltext extraction tool to be run first. Its performance is quite good and seems to have been improving recently (cf. Tkaczyk et al. [57] vs Cioffi and Peroni [7]), making it a relatively new alternative to Cermin and Grobid.

4 Experiments

We evaluate all toolchains against three gold standards as described in the following section. We separate the performance by the following fields that any reference parser will commonly identify: reference string, title, publication year, author names, editor names, publisher, source (e.g. journal or collection), volume and issue (if journal article), start-page and end-page (in the source). Reporting results separately for different fields allows the reader to identify which tools perform best for the fields most important to the user. Performance on the reference string as such evaluates extraction without parsing. Extraction of global identifiers like URL, DOI, or arxiv-id

Table 2: Input and output options of the tools.

INPUT				TOOL	OUTPUT				
•			• ⁵	Grobid	•	•	•	•	•
•			•	Cermin	•	•	•	•	•
		•	•	AnyStyle			•	•	
				ExParser				•	
•				pdftotext	•				
pdf	layout	text	refstr		layout	text	refstr	refobj	meta

⁵According to [58], Grobid can be used to parse reference strings, but we did not see how as noted for Grobid in Table 2.

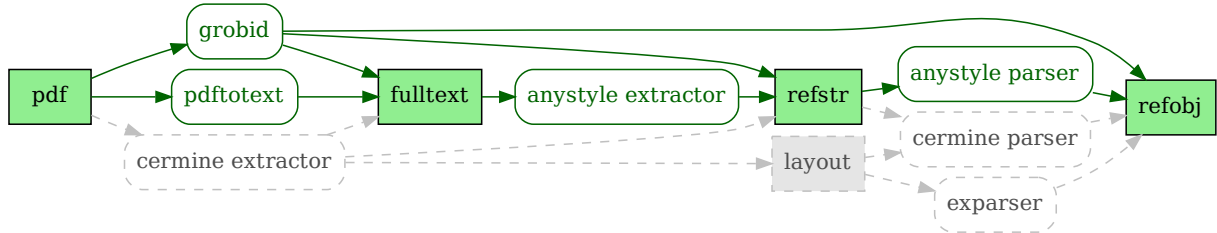


Fig. 1: The toolchains evaluated as best performing in green, and the noncompetitive ones greyed out.

would be desirable, but the different tools handle these differently: AnyStyle extracts DOIs, Grobid extracts arxiv-ids, and neither Cermine nor Exparsar extracts any of them. However, in our framework, we have verified that it is easy to use regular expressions and URL resolution to extract such identifiers from the reference string in a post-processing step.

4.1 Datasets

We use three datasets of manually extracted and parsed references that have been created in the EXCITE⁶ project [27], in the GEOCITE⁷ project [2], and by Cioffi and Peroni [7], respectively⁸. The assumption is that the tools that best approximate the gold standards are the best-performing tools. The EXCITE dataset contains 348 full-text PDFs with metadata from the SSOAR⁹ repository. The GEOCITE dataset contains 178 full-text PDFs from the geosciences domain. Both contain German and English documents and scans that require OCR. The CIOFFI dataset is described in Cioffi and Peroni [7] and contains 56 relatively

recent English scientific papers sampled from various domains. Table 3 presents some statistics about the number of annotated references.

4.2 Evaluation measures

To evaluate different extraction and parsing tools, we compare the parsed reference objects against the reference objects as provided in the gold standard. This means we compare two nested objects. Requiring the parsers to reproduce exactly the gold reference object would make the task too difficult. Allowing irrelevant minor differences, we want the parsers to approximate the gold object as much as possible. If a parser approximates a gold object to some extent, this should also count towards its performance score. The attribute-value matching done for evaluation is illustrated in Figure 2. As a first simplification, we assume that both gold and parsed reference objects are sets of attribute-value pairs. This ignores nestedness (for example, it does not matter if a surname belongs to the first or second author) and order (e.g. of authors). So it will suffice if all the surnames are found, all the first names are found, etc. We consider it unlikely that a tool systematically swaps name parts between different authors or changes the order of authors mentioned, so these relaxed requirements should not unjustly favour some tools over others, in contrast to overly restrictive requirements that can indeed arbitrarily favour certain tools based on irrelevant minor differences.¹⁰ Further, we assume that the values in these attribute-value pairs need not match exactly but only closely (for example, additional whitespace or capitalization should be ignored).

⁶<https://github.com/exciteproject>

⁷<https://github.com/GeoCite>

⁸As the gold standard by Cioffi and Peroni [7] does not include the extracted, unparsed refstrings, we had to create artificial ones from the provided parsed refojects, assuming that these will be sufficiently similar to the original refstrings.

⁹www.ssoar.info

Table 3: Number of references per document in the EXCITE, GEOCITE and CIOFFI datasets.

	# Docs	# Refs	Mean	Lowest	Highest	0-19	20-39	40-59	60-79	80-99	100-119	120-139	140-159
GEOCITE	178	4729	27	0	109	86	48	29	7	5	3	0	0
EXCITE	348	8679	25	1	151	162	111	63	8	2	0	1	1
CIOFFI	56	2536	45	10	113	5	23	14	9	4	1	0	0

¹⁰Consider the case of the BLEU score [45] in machine translation, which under normal assumptions will measure translation quality well, although it is possible to intentionally produce a bad translation with a good score.

reference	Abeshouse, Bob, The Koch Brothers -People & Power: Al Jazeera English. N.p., 1 Nov., 2011. http://www.aljazeera.com/programmes/peopleandpower/2011/10/2011102683719370179 .html [Accessed on 16 Nov. 2011.]										
	[1] Abeshouse, Bob, The Koch Brothers - People & Power: Al Jazeera English. N.p., 1 Nov., 2011. http://www.aljazeera.com/programmes/peopleandpower/2011/10/2011102683719370179 .html [Accessed on 16 Nov. 2011.]										
	2011	Al Jazeera English. N.p	author	Bob	Abeshouse	The	Koch	Brothers	People	Power	
	2011			Bob	Abeshouse						The Koch Brothers - People & Power: Al Jazeera English
reference	Vgl. Engel, Ulf (2005) : Deutschland, Afrika und die Entstehung gemeinsamer Interessen, in: Aus Politik										
	Engel, Ulf (2005): Deutschland, Afrika und die Entstehung gemeinsamer Interessen, in: Aus Politik und Zeitgeschichte 4/2005, 11–17.										
	2005		author	Ulf	Engel	Vgl	Aus Politik	in	: Deutschland, Afrika und die Entstehung gemeinsamer Interessen	volume	4
	2005	Aus Politik und Zeitgeschichte		Ulf	Engel				Deutschland, Afrika und die Entstehung gemeinsamer Interessen	start	11
										end	17

Fig. 2: Two visualized examples of matchings between aligned system (blue, Positive) and gold (yellow, True) references. Green indicates a match (True-Positive). We count the coloured boxes towards TP, P and T to compute precision as $\frac{TP}{P} = \frac{4}{10}$ (top) or $\frac{TP}{P} = \frac{5}{8}$ (bottom) and recall as $\frac{TP}{T} = \frac{4}{5}$ (top) or $\frac{TP}{T} = \frac{5}{6}$ (bottom). Note that in our evaluation, we count TP, T and P over all references of a given document and separate the counts by fields (for example, ‘author’). The final results are the average over the individual documents’ precision and recall.

Consider that we are evaluating a parsing problem, which is mostly about sequence labelling. Mistakes in one part of the parsing will affect other parts. This means that it is unlikely that the accepted differences will grant an unfair advantage to some tools, as even an accepted mistake will correspond to further unaccepted mistakes in other parts. We require that a maximally different parse has a very low score, a perfect parse has a perfect score, and anything in between gives a reasonable fine-grained impression of how correct the parse is. The more wrong a parse is, the worse should be the score. If a is more wrong than b , then the score of a should be lower than that of b . We argue this is achieved as follows: After the attribute-value pairs are obtained, they are grouped by attributes. Then, for each attribute, the optimal pairing between gold and reference values is computed based on a string similarity measure (in our case, the Python *difflib Sequence-Matcher* based on the algorithm by Ratcliff et al. [50] for strings and Boolean equality for other values like numbers) and the *linear sum assignment* or *minimum weight matching*. This *unbalanced assignment problem* [49] is a fairly expensive combinatorial optimization, but since the compared sets of values for a given attribute are quite small, it is feasible in practice. We assume that alignment

of two values for the same attribute is necessary but not sufficient for them to match. If not sufficiently similar, values may still be aligned by the above algorithm for lack of better alternatives. After the above described alignment, a threshold on the Ratcliff et al. [50] alignment similarity or possibly any other string similarity measure is applied to determine whether the aligned values are sufficiently similar to be considered a match. In our matching, the alignment similarity was divided by the number of characters of the shorter aligned value in case of most string attributes and a threshold was roughly tuned manually by inspection in a reference-to-metadata-collection matching scenario on different data.¹¹ For year attributes, we allow a maximal integer difference of 1 between the values.¹² For all other attributes, we require exact value match. Trues (T) is the number of elements in the gold-standard object. Positives (P) is the number of elements in the extracted object. And true-positives (TP) is the number of matching element-pairs. Then precision and recall are computed as TP/P and TP/T ,

¹¹We consider it unlikely that this threshold’s fine-tuning favours some tools over others.

¹²While this matters in other matching applications, here it essentially amounts to exact match of extracted and true year.

respectively. As multiple extracted and gold reference objects exist for each document, we apply the assignment procedure first to extracted vs. gold reference strings before continuing to compare their parsed versions as described above.

5 Results

Results of our experiments are displayed in Table 4, showing precision, recall and F1 of different toolchains on the GEOCITE, EXCITE and CIOFFI datasets for fields commonly extracted by those toolchains, like *title* or *authors* as well as pure reference string extraction in the first column under ‘*refstr*’. The corresponding Table 5 specifies the best performance for each column in violet and all other performances in magenta as a percentage drop (not percentage *point* drop!) of the best performance. For example, *anystyle references from cermine fulltext’s title* performance of 69% F1 is 16% lower than the best performance, which is by

anystyle references from grobid refstrings. Naturally, *grobid references from grobid xml*, *cermine references from grobid refstrings* and *anystyle references from grobid refstrings* all have the same reference string extraction performance because they use the grobid-extracted reference strings.

5.1 Main observations

As Table 4 presents a great number of individual evaluation results and different users might be interested in different fields, in the following, we will refrain from describing the entire table and instead focus on summarizing the bottom line from our perspective, not giving priority to any specific attributes. On the GEOCITE dataset, the best-performing toolchain is the AnyStyle parser applied on Grobid-extracted reference strings. It has the best F1 of all tools in 10 out of 11 fields, including reference extraction (*refstr*). Interestingly, this ranking is slightly different for

Table 4: Performance predicting the most important reference fields in GEOCITE, EXCITE and CIOFFI. Evaluation results in terms of precision, F1 and recall. Best performing tool per field in **violet**.

performance	P	F1	R	P	F1	R	P	F1	R	P	F1	R	P	F1	R	P	F1	R	P	F1	R	P	F1	R	P	F1	R	P	F1	R			
GEOCITE	refstr			title			year			author			editor			publ			source			vol			issue			startp			endp		
cerm.txt–anyst	95	77	65	86	69	58	93	73	62	65	62	60	47	32	26	22	19	18	69	59	52	70	57	48	41	36	33	72	63	58	76	65	58
cerm.ref–anyst	90	83	77	85	75	68	91	80	73	64	67	70	48	38	32	24	21	21	67	61	56	71	61	54	41	37	34	74	70	67	81	72	66
grob.txt–anyst	94	86	80	89	81	75	94	82	73	66	71	77	58	45	38	26	24	25	71	68	66	74	63	56	43	41	39	80	77	74	84	78	73
grob.ref–anyst	90	87	83	89	83	78	92	84	78	67	73	80	57	46	39	26	25	27	72	69	67	78	66	58	47	44	43	80	78	76	84	79	75
pdftotxt–anyst	95	81	72	87	74	65	91	77	67	64	65	67	49	39	33	24	21	22	70	63	58	72	59	51	47	42	38	74	68	64	77	69	62
cermine	90	83	77	80	68	60	90	78	69	56	57	59	0	–	0	0	–	0	47	37	32	57	50	46	29	22	19	57	52	49	73	59	49
grob.ref–cerm	90	87	83	89	79	72	91	82	75	69	60	54	0	–	0	0	–	0	52	45	41	58	53	50	31	27	26	61	57	56	76	64	56
cerm.layo–exp	88	74	65	84	70	61	85	69	58	59	57	55	45	34	28	16	11	11	58	52	48	54	50	46	40	37	36	58	52	48	60	51	45
grobid	90	87	83	86	78	71	89	82	78	71	75	81	47	26	18	25	18	17	67	56	49	73	65	58	37	29	24	82	78	74	83	78	74
EXCITE	refstr			title			year			author			editor			publ			source			vol			issue			startp			endp		
cerm.txt–anyst	93	85	78	90	81	74	93	84	76	82	79	76	67	56	50	52	50	49	72	67	63	73	62	55	46	44	44	77	71	65	80	70	64
cerm.ref–anyst	89	82	76	85	78	73	90	81	75	78	76	75	60	52	46	47	45	44	71	67	64	69	61	55	42	38	37	75	71	67	79	72	68
grob.txt–anyst	87	84	82	83	79	77	89	83	79	75	77	80	65	58	53	49	49	50	65	66	68	72	63	56	44	43	43	76	74	73	80	77	74
grob.ref–anyst	83	85	87	81	81	82	88	85	83	75	79	84	63	57	53	49	51	53	66	67	69	71	64	58	45	44	45	75	75	74	79	77	75
pdftotxt–anyst	90	87	85	88	84	81	90	86	82	79	81	83	67	59	53	50	51	53	71	69	69	70	63	58	43	43	44	77	75	73	80	75	72
cermine	89	82	76	81	71	63	89	80	73	66	61	57	0	–	0	0	–	0	53	46	41	61	55	50	36	31	28	63	60	57	77	63	55
grob.ref–cerm	83	85	87	80	77	74	88	84	81	69	58	50	0	–	0	0	–	0	50	46	43	58	56	55	39	36	34	65	63	61	75	67	61
cerm.layo–exp	87	79	72	85	77	71	87	75	66	74	67	62	59	55	53	50	44	40	65	62	59	57	56	56	43	42	42	59	56	54	61	57	53
grobid	83	85	87	77	77	76	84	83	82	74	78	84	51	33	25	47	41	37	57	51	47	68	63	60	39	32	28	74	74	75	75	73	73
CIOFFI	refstr			title			year			author			editor			publ			source			vol			issue			startp			endp		
cerm.txt–anyst	91	83	77	89	80	74	90	82	75	61	66	74	11	15	28	53	52	52	74	72	72	85	77	72	–	–	–	84	75	67	83	73	66
cerm.ref–anyst	94	90	88	91	87	84	92	89	85	63	72	86	12	17	33	52	52	52	72	76	81	88	85	81	–	–	–	87	83	80	82	79	76
grob.txt–anyst	93	91	91	92	90	89	91	90	88	65	75	89	19	23	35	63	57	54	81	85	88	89	88	88	–	–	–	88	85	83	89	85	83
grob.ref–anyst	93	93	93	92	92	92	92	92	92	66	76	91	19	24	36	57	53	52	82	86	91	90	90	91	–	–	–	88	86	85	89	87	85
pdftotxt–anyst	91	85	81	86	81	77	88	82	77	58	65	76	9	12	28	54	52	53	74	73	74	83	79	76	–	–	–	84	76	71	82	74	70
cermine	94	90	88	90	83	77	92	88	84	62	69	79	0	–	0	0	–	0	68	70	72	87	84	83	–	–	–	88	85	83	85	81	77
grob.ref–cerm	93	93	93	93	91	90	91	91	90	74	69	66	0	–	0	0	–	0	79	82	85	89	89	90	–	–	–	89	87	85	88	87	86
cerm.layo–exp	92	83	77	88	81	75	84	73	65	56	60	65	6	4	14	41	36	32	73	72	72	65	63	61	–	–	–	77	71	67	70	66	63
grobid	93	93	93	91	91	91	92	92	92	67	77	92	16	14	17	57	51	48	83	85	88	90	89	89	–	–	–	90	91	92	87	87	87

EXCITE, where AnyStyle applied on pdftotext fulltext has the most (8/11 incl. *refstr*) top-F1 columns. However, for those fields where AnyStyle on pdftotext performs better than AnyStyle on Grobid-extracted reference strings, the latter is only slightly worse (1-3%). On CIOFFI, again AnyStyle on Grobid performs best, with 7/11 incl. *refstr*. This shows that AnyStyle is clearly useful, whether applied on Grobid-extracted reference strings or on pdftotext fulltext. Tools other than AnyStyle or Grobid are not competitive according to our comparison (see also Figure 1). Cerminе performs worst, while ExParser, which uses Cerminе-extracted layouts but not Cerminе-extracted reference strings, performs better for some features – and worse for others. Cerminе seems not to parse editors and publishers. The performance of most tools’ reference extraction is similar to the performance of their parsing, except for ExParser, which has particularly poor performance here. On the contrary, the worst part of

Cerminе is its parser (compare AnyStyle references from Cerminе reference strings to Cerminе end-to-end). However, AnyStyle also works better with the basic pdftotext fulltext input than with Cerminе or Grobid full-text input. Overall, the best choice is to use AnyStyle parser and Grobid-extracted references.

5.2 Comparing our Evaluation

Comparing our results on the CIOFFI dataset against those on GEOCITE and EXCITE, as well as on those reported in Cioffi and Peroni [7], we note that our CIOFFI evaluation is very similar to our GEOCITE evaluation – except for *AnyStyle references from Cerminе fulltext* vs. *AnyStyle references from Cerminе references* regarding reference string and year. The original results reported by Cioffi and Peroni [7] are different, in particular the poor performance of Grobid reference extraction surprises, given that Grobid provides some of

Table 5: Percentage-drop (magenta; not %-point-drop!) from best performance (violet) in Table 4. For example, refstr-F1 is 4% lower than the best result (87) for *cerminе refs*→*anystyle* on GEOCITE.

% drop	P F1 R	P F1 R	P F1 R	P F1 R	P F1 R	P F1 R	P F1 R	P F1 R	P F1 R	P F1 R	P F1 R	P F1 R
GEOCITE	refstr	title	year	author	editor	publ	source	vol	issue	startp	endp	
cerm.txt–anyst	95 11 21	3 16 25	1 13 20	8 17 25	18 30 33	15 24 33	4 14 22	10 13 17	12 18 23	12 19 23	9 17 22	
cerm.ref–anyst	5 4 7	4 9 12	3 4 6	9 10 13	17 17 17	7 16 22	6 11 16	8 7 6	12 15 20	9 10 11	3 8 12	
grob.txt–anyst	1 1 3	89 2 3	94 2 6	7 5 4	57 2 2	26 4 7	1 1 1	5 4 3	8 6 9	2 1 2	84 1 2	
grob.ref–anyst	5 87 83	89 83 78	2 84 78	5 2 1	1 46 39	26 25 27	72 69 67	78 66 57	47 44 43	2 78 76	84 79 75	
pdftotxt–anyst	95 6 13	2 10 16	3 8 14	9 13 17	15 15 15	7 16 18	2 8 13	7 10 12	47 4 11	9 12 15	8 12 17	
cerminе	5 4 7	10 18 23	4 7 11	21 24 27	– – –	– – –	34 46 52	26 24 20	38 50 55	30 33 35	13 25 34	
grob.ref–cerm	5 87 83	89 4 7	3 2 3	2 19 33	– – –	– – –	27 34 38	25 19 13	34 38 39	25 26 26	9 18 25	
cerm.layo–exp	7 14 21	5 15 21	9 17 25	16 24 32	22 26 28	38 56 59	19 24 28	30 24 20	14 15 16	29 33 36	28 35 40	
grobid	5 87 83	3 6 8	5 2 78	71 75 81	18 43 53	3 28 37	6 18 26	6 1 57	21 34 44	82 78 2	1 1 1	
EXCITE	refstr	title	year	author	editor	publ	source	vol	issue	startp	endp	
cerm.txt–anyst	93 2 10	90 3 9	93 2 8	82 2 9	67 5 5	52 1 7	72 2 8	73 3 8	46 44 2	77 5 13	80 9 14	
cerm.ref–anyst	4 5 12	5 7 10	3 5 9	4 6 10	10 11 13	9 11 16	1 2 7	5 4 8	8 13 17	2 5 10	1 6 9	
grob.txt–anyst	6 3 5	7 5 6	4 3 4	8 4 4	2 1 53	5 3 5	9 4 1	1 1 6	4 2 4	1 1 2	80 77 1	
grob.ref–anyst	10 2 87	9 3 82	5 1 83	8 2 84	5 3 53	5 51 53	8 2 69	2 64 3	2 44 45	2 75 1	1 77 75	
pdftotxt–anyst	3 87 2	2 84 1	3 86 1	3 81 1	67 59 53	3 51 53	1 69 69	4 1 3	6 2 2	77 75 2	80 2 4	
cerminе	4 5 12	9 15 23	4 6 12	19 24 32	– – –	– – –	26 33 40	16 14 16	21 29 37	18 19 24	3 18 26	
grob.ref–cerm	10 2 87	11 8 9	5 2 2	15 28 40	– – –	– – –	30 33 37	20 12 8	15 18 24	15 16 18	6 12 18	
cerm.layo–exp	6 9 17	5 8 13	6 12 20	9 17 26	11 6 53	3 13 24	9 10 14	21 12 6	6 4 6	23 25 28	23 25 29	
grobid	10 2 87	14 8 7	9 3 1	9 3 84	23 44 52	9 19 30	20 26 31	6 1 60	15 27 37	3 1 75	6 5 2	
CIOFFI	refstr	title	year	author	editor	publ	source	vol	issue	startp	endp	
cerm.txt–anyst	3 10 17	4 13 19	2 10 18	17 14 19	42 37 22	15 8 3	10 16 20	5 14 20	– – –	6 17 27	6 16 24	
cerm.ref–anyst	94 3 5	2 5 8	92 3 7	14 6 6	36 29 8	17 8 3	13 11 10	2 5 10	– – –	3 8 13	7 9 12	
grob.txt–anyst	1 2 2	1 2 3	1 2 4	12 2 3	19 4 2	63 56 54	2 1 3	1 2 3	– – –	2 6 9	89 2 4	
grob.ref–anyst	1 93 93	1 92 92	92 92 92	10 1 1	19 24 36	9 7 3	1 86 91	90 90 91	– – –	2 5 7	89 87 2	
pdftotxt–anyst	3 8 12	7 11 16	4 10 16	21 15 17	52 50 22	14 8 1	10 15 18	7 12 16	– – –	6 16 22	7 14 19	
cerminе	94 3 5	3 9 16	92 4 8	16 10 14	– – –	– – –	18 18 20	3 6 8	– – –	2 6 9	4 6 11	
grob.ref–cerm	1 93 93	93 1 2	1 1 2	74 10 28	– – –	– – –	4 4 6	1 1 1	– – –	1 4 7	1 87 1	
cerm.layo–exp	2 10 17	5 11 18	8 20 29	24 22 29	68 83 61	34 36 40	12 16 20	27 30 32	– – –	14 21 27	21 24 27	
grobid	1 93 93	2 1 1	92 92 92	9 77 92	15 41 52	9 10 11	83 1 3	90 1 2	– – –	90 91 92	2 87 87	

✓ Parsed references via Anystyle using Grobid refstrings

1. [book] **"Knowledge work and knowledge-intensive firms"**
M Alvesson (2004)
Oxford University Press, Oxford, UK
SEARCH ON WEB: <GOOGLE> <GOOGLE SCHOLAR> <BING>
2. [article-journal] **"Identity Regulation as Organizational Control: Producing the Appropriate Individual"**
M Alvesson, H Willmott (2002)
Journal of Management Studies, 39, (5), pp. 619 - 644
CROSSREF: [10.1111/1467-6486.00305](https://doi.org/10.1111/1467-6486.00305) **OPENALEX:** <https://openalex.org/W3123980981> **URL:** onlinelibrary.wiley.com

Fig. 3: One toolchain with two extracted, parsed and linked references in the OUTCITE demonstrator.

the best reference extraction in our experiments. Their results align better with our overall results for parsing (metadata and content) evaluation, although they do not provide results for individual fields. In general, from their description, it seems that their evaluation is based more on identifying expected spans in the provided text compared to extracting substrings mostly identical to the expected values as in our case, which explains the result divergence (for example, some tools extract a reference number before the reference, while others do not, which could be considered a mistake in their evaluation, but not in ours).

6 Conclusion

We have conducted an empirical evaluation of different toolchains applied to PDF publications from the social sciences (EXCITE), geosciences (GEOCITE) and a broader mix of domains (CIOFFI). These publications had been manually processed (not by us) to extract and parse their references, which allows for reliable evaluation of the compared tools' performance. We have distinguished the task of extracting only the reference string and parsing a reference string into a reference object, i.e., a set of meta-information on the referenced publication. When comparing several existing tools in different combinations, we find that a subset consisting of pdftotext, Grobid and AnyStyle is relevant. We expect that when tweaking the full-text-extraction from Grobid-extracted XML, one can do better than pdftotext, leaving just Grobid and AnyStyle. One should also note that Cermin is quite slow compared to Grobid. Our comparison enables users interested in particular fields to make their own choices based on the individual performance reported for those. Based

on our evaluation, we recommend using AnyStyle for parsing and Grobid for extraction.

Resources

For testing the toolchains evaluated in this paper, we provide a demo system at <https://demo-outcite.gesis.org/> which shows the reference extraction and parsing results of uploaded PDF files (see Figure 3). Further, we provide the code for our framework at <https://github.com/OUTCITE/outcite-refextract> or <https://archive.softwareheritage.org/browse/directory/b8b3db04fb8212d3b5cda0f45a2e86bc57833257/> and under *doi*: 10.5281/zenodo.10985784, which includes the evaluation script as well as the gold data. The authors are willing to help interested readers navigate our repositories and apply our framework to their systems to foster dissemination, exploitation and reproducibility.

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References

- [1] Dhananjay Ashok and Zachary C. Lipton. PromptNER: Prompting For Named Entity Recognition, June 2023. URL <http://arxiv.org/abs/2305.15444>. arXiv:2305.15444 [cs].

- [2] Bastian Birkeneder, Philipp Aufenvenne, Christian Haase, Philipp Mayr, and Malte Steinbrink. Extracting literature references in German Speaking Geography – the GEOcite project. In *Proceedings of the Workshop on Understanding Literature references in academic full TExt*, pages 34–41. CEUR-WS.org, 2022. URL <http://ceur-ws.org/Vol-3220/paper4.pdf>.
- [3] Timo Borst, Jonas Mielck, Matthias Nannt, and Wolfgang Riese. Extracting Funder Information from Scientific Papers - Experiences with Question Answering. In Gianmaria Silvello, Oscar Corcho, Paolo Manghi, Giorgio Maria Di Nunzio, Koraljka Golub, Nicola Ferro, and Antonella Poggi, editors, *Linking Theory and Practice of Digital Libraries*, volume 13541, pages 289–296. Springer International Publishing, Cham, 2022. ISBN 978-3-031-16801-7 978-3-031-16802-4. doi: 10.1007/978-3-031-16802-4_24. URL https://link.springer.com/10.1007/978-3-031-16802-4_24. Series Title: Lecture Notes in Computer Science.
- [4] Zeyd Boukhers, Shriharsh Ambhore, and Steffen Staab. An end-to-end approach for extracting and segmenting high-variance references from pdf documents. In *2019 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, pages 186–195, 2019. doi: 10.1109/JCDL.2019.00035.
- [5] Salvatore Carta, Alessandro Giuliani, Leonardo Piano, Alessandro Sebastian Podda, Livio Pompianu, and Sandro Gabriele Tiddia. Iterative Zero-Shot LLM Prompting for Knowledge Graph Construction. *arXiv*, July 2023. URL <http://arxiv.org/abs/2307.01128>. arXiv:2307.01128 [cs].
- [6] Chien-Chih Chen, Kai-Hsiang Yang, Chuen-Liang Chen, and Jan-Ming Ho. Bibpro: A citation parser based on sequence alignment. *IEEE Transactions on Knowledge and Data Engineering*, 24(2):236–250, 2010.
- [7] Alessia Cioffi and Silvio Peroni. Structured references from pdf articles: Assessing the tools for bibliographic reference extraction and parsing. In *Linking Theory and Practice of Digital Libraries: 26th International Conference on Theory and Practice of Digital Libraries, TPDL 2022, Padua, Italy, September 20–23, 2022, Proceedings*, page 425–432, Berlin, Heidelberg, 2022. Springer-Verlag. ISBN 978-3-031-16801-7. doi: 10.1007/978-3-031-16802-4_42. URL https://doi.org/10.1007/978-3-031-16802-4_42.
- [8] Giovanni Colavizza and Matteo Romanello. Annotated references in the historiography on venice: 19th–21st centuries. *Journal of Open Humanities Data*, Nov 2017. doi: 10.5334/johd.9.
- [9] Alexandru Constantin, Steve Pettifer, and Andrei Voronkov. Pdfx: fully-automated pdf-to-xml conversion of scientific literature. In *Proceedings of the 2013 ACM symposium on document engineering*, pages 177–180, 2013.
- [10] Eli Cortez, Altigran S da Silva, Marcos André Gonçalves, Filipe Mesquita, and Edleno S de Moura. Flux-cim: flexible unsupervised extraction of citation metadata. In *Proceedings of the 7th ACM/IEEE-CS joint conference on Digital libraries*, pages 215–224, 2007.
- [11] Eli Cortez, Altigran S da Silva, Marcos André Gonçalves, Filipe Mesquita, and Edleno S de Moura. A flexible approach for extracting metadata from bibliographic citations. *Journal of the American Society for Information Science and Technology*, 60(6):1144–1158, 2009.
- [12] Isaac G Council, C Lee Giles, and Min-Yen Kan. Parscit: an open-source crf reference string parsing package. In *LREC*, volume 8, pages 661–667, 2008.
- [13] Nguyen Viet Cuong, Muthu Kumar Chandrasekaran, Min-Yen Kan, and Wee Sun Lee. Scholarly document information extraction using extensible features for efficient higher order semi-crfs. In *Proceedings of the 15th ACM/IEEE-CS joint conference on digital libraries*, pages 61–64, 2015.

- [14] John Dagdelen, Alexander Dunn, Sanghoon Lee, Nicholas Walker, Andrew S Rosen, Gerbrand Ceder, Kristin A Persson, and Anubhav Jain. Structured information extraction from scientific text with large language models. *Nature Communications*, 15(1):1418, 2024.
- [15] Marilena Daquino, Silvio Peroni, David Shotton, Giovanni Colavizza, Behnam Ghavimi, Anne Lauscher, Philipp Mayr, Matteo Romanello, and Philipp Zumstein. The OpenCitations Data Model. In *Proceedings of International Semantic Web Conference (ISWC 2020)*, pages 447–463. Springer, 2020. doi: 10.1007/978-3-030-62466-8_28. URL <https://arxiv.org/abs/2005.11981>.
- [16] Min-Yuh Day, Richard Tzong-Han Tsai, Cheng-Lung Sung, Chiu-Chen Hsieh, Cheng-Wei Lee, Shih-Hung Wu, Kun-Pin Wu, Chorng-Shyong Ong, and Wen-Lian Hsu. Reference metadata extraction using a hierarchical knowledge representation framework. *Decision Support Systems*, 43(1):152–167, 2007.
- [17] Eugene Garfield. Citation indexes for science. *Science*, 122(3159):108–111, 1955. doi: 10.1126/science.122.3159.108. URL <https://www.science.org/doi/abs/10.1126/science.122.3159.108>.
- [18] Zoubin Ghahramani. An introduction to hidden markov models and bayesian networks. *International journal of pattern recognition and artificial intelligence*, 15(01):9–42, 2001.
- [19] Akshay Goel, Almog Gueta, Omry Gilon, Chang Liu, Sofia Erell, Lan Huong Nguyen, Xiaohong Hao, Bolous Jaber, Shashir Reddy, Rupesh Kartha, et al. Llms accelerate annotation for medical information extraction. In *Machine Learning for Health (ML4H)*, pages 82–100. PMLR, 2023.
- [20] Mark Grennan and Joeran Beel. Synthetic vs. real reference strings for citation parsing, and the importance of re-training and out-of-sample data for meaningful evaluations: Experiments with GROBID, GIANT and CORA. In Petr Knuth, Christopher Stahl, Bikash Gyawali, David Pride, Suchetha N. Kunnath, and Drahomira Herrmannova, editors, *Proceedings of the 8th International Workshop on Mining Scientific Publications*, pages 27–35, Wuhan, China, 05 August 2020. Association for Computational Linguistics. URL <https://aclanthology.org/2020.wosp-1.4>.
- [21] Mark Grennan, Martin Schibel, Andrew Collins, and Joeran Beel. Giant: The 1-billion annotated synthetic bibliographic-reference-string dataset for deep citation parsing. In *27th AIAI Irish Conference on Artificial Intelligence and Cognitive Science*, pages 101–112, 2019.
- [22] Tudor Groza, AAstrand Grimnes, and Siegfried Handschuh. Reference information extraction and processing using random conditional fields. *Information Technology and Libraries*, 31(2):6–20, 2012.
- [23] Zhixin Guo and Hai Jin. Reference meta-data extraction from scientific papers. In *2011 12th International Conference on Parallel and Distributed Computing, Applications and Technologies*, pages 45–49. IEEE, 2011.
- [24] Deepank Gupta, Bob Morris, Terry Catapano, and Guido Sautter. A new approach towards bibliographic reference identification, parsing and inline citation matching. In *International Conference on Contemporary Computing*, pages 93–102. Springer, 2009.
- [25] Muhammad Usman Hadi, Rizwan Qureshi, Abbas Shah, Muhammad Irfan, Anas Zafar, Muhammad Bilal Shaikh, Naveed Akhtar, Jia Wu, Seyedali Mirjalili, et al. A survey on large language models: Applications, challenges, limitations, and practical usage. *Authorea Preprints*, 2023.
- [26] Erik Hetzner. A simple method for citation metadata extraction using hidden markov models. In *Proceedings of the 8th ACM/IEEE-CS joint conference on Digital libraries*, pages 280–284, 2008.
- [27] Azam Hosseini, Behnam Ghavimi, Zeyd Boukhers, and Philipp Mayr. Excite –

- a toolchain to extract, match and publish open literature references. In *2019 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, pages 432–433, 2019. doi: 10.1109/JCDL.2019.00105.
- [28] Yu-Lun Hsieh, Shih-Hung Liu, Ting-Hao Yang, Yu-Hsuan Chen, Yung-Chun Chang, Gladys Hsieh, Cheng-Wei Shih, Chun-Hung Lu, and Wen-Lian Hsu. A frame-based approach for reference metadata extraction. In *International Conference on Technologies and Applications of Artificial Intelligence*, pages 154–163. Springer, 2014.
 - [29] B. Ian Hutchins. A tipping point for open citation data. *Quantitative Science Studies*, 2(2):433–437, 07 2021. ISSN 2641-3337. doi: 10.1162/qss.c.00138. URL <https://doi.org/10.1162/qss.c.00138>.
 - [30] Ariani Indrawati, Ambar Yoganingrum, and Pradipta Yuwono. Evaluating the quality of the indonesian scientific journal references using parscit, cermin and grobid. *Library Philosophy and Practice*, pages 1–14, 2019.
 - [31] Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
 - [32] Roman Kern, Kris Jack, Maya Hristakeva, and Michael Granitzer. Teambeam-metadata extraction from scientific literature. *D-Lib Magazine*, 18(7):1, 2012.
 - [33] Young-Min Kim, Patrice Bellot, Jade Tavernier, Elodie Faath, and Marin Dacos. Evaluation of bilbo reference parsing in digital humanities via a comparison of different tools. In *Proceedings of the 2012 ACM symposium on Document engineering*, pages 209–212, 2012.
 - [34] Martin Körner, Behnam Ghavimi, Philipp Mayr, Heinrich Hartmann, and Steffen Staab. Evaluating reference string extraction using line-based conditional random fields: A case study with german language publications. In Mārīte Kirikova, Kjetil Nørvåg, George A. Papadopoulos, Johann Gamper, Robert Wrembel, Jérôme Darmont, and Stefano Rizzi, editors, *New Trends in Databases and Information Systems*, pages 137–145, Cham, 2017. Springer International Publishing. ISBN 978-3-319-67162-8.
 - [35] John D. Lafferty, Andrew McCallum, and Fernando C. N. Pereira. Conditional random fields: Probabilistic models for segmenting and labeling sequence data. In *Proceedings of the Eighteenth International Conference on Machine Learning, ICML ’01*, page 282–289, San Francisco, CA, USA, 2001. Morgan Kaufmann Publishers Inc. ISBN 1558607781.
 - [36] Peder Larsen and Markus Von Ins. The rate of growth in scientific publication and the decline in coverage provided by science citation index. *Scientometrics*, 84(3):575–603, 2010.
 - [37] Bo Li, Gexiang Fang, Yang Yang, Quansen Wang, Wei Ye, Wen Zhao, and Shikun Zhang. Evaluating ChatGPT’s Information Extraction Capabilities: An Assessment of Performance, Explainability, Calibration, and Faithfulness, April 2023. URL <http://arxiv.org/abs/2304.11633>. arXiv:2304.11633 [cs].
 - [38] Mario Lipinski, Kevin Yao, Corinna Breiteringer, Joeran Beel, and Bela Gipp. Evaluation of header metadata extraction approaches and tools for scientific pdf documents. In *Proceedings of the 13th ACM/IEEE-CS Joint Conference on Digital Libraries, JCDL ’13*, page 385–386, New York, NY, USA, 2013. Association for Computing Machinery. ISBN 9781450320771. doi: 10.1145/2467696.2467753. URL <https://doi.org/10.1145/2467696.2467753>.
 - [39] Patrice Lopez. Grobid: Combining automatic bibliographic data recognition and term extraction for scholarship publications. In *International conference on theory and practice of digital libraries*, pages 473–474. Springer, 2009.

- [40] Daiki Namikoshi, Manabu Ohta, Atsuhiko Takasu, and Jun Adachi. Crf-based bibliography extraction from reference strings using a small amount of training data. In *2017 Twelfth International Conference on Digital Information Management (ICDIM)*, pages 59–64. IEEE, 2017.
- [41] Manabu Ohta, Daiki Arauchi, Atsuhiko Takasu, and Jun Adachi. Crf-based bibliography extraction from reference strings focusing on various token granularities. In *2012 10th IAPR International Workshop on Document Analysis Systems*, pages 276–281. IEEE, 2012.
- [42] Manabu Ohta, Daiki Arauchi, Atsuhiko Takasu, and Jun Adachi. Empirical evaluation of crf-based bibliography extraction from reference strings. In *2014 11th IAPR International Workshop on Document Analysis Systems*, pages 287–292. IEEE, 2014.
- [43] Bolanle Ojokoh, Ming Zhang, and Jian Tang. A trigram hidden markov model for meta-data extraction from heterogeneous references. *Information Sciences*, 181(9):1538–1551, 2011.
- [44] Olga Pagnotta. Reflex project - dataset and gold standard, January 2024. URL <https://doi.org/10.5281/zenodo.10524830>.
- [45] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In Pierre Isabelle, Eugene Charniak, and Dekang Lin, editors, *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, Philadelphia, Pennsylvania, USA, July 2002. Association for Computational Linguistics. doi: 10.3115/1073083.1073135. URL <https://aclanthology.org/P02-1040>.
- [46] Fuchun Peng and Andrew McCallum. Information extraction from research papers using conditional random fields. *Information processing & management*, 42(4):963–979, 2006.
- [47] Silvio Peroni and David Shotton. Opencitations, an infrastructure organization for open scholarship. *Quantitative Science Studies*, 1(1):428–444, 2020.
- [48] Animesh Prasad, Manpreet Kaur, and Min-Yen Kan. Neural parsit: a deep learning-based reference string parser. *International journal on digital libraries*, 19(4):323–337, 2018.
- [49] Lyle Ramshaw and Robert E Tarjan. On minimum-cost assignments in unbalanced bipartite graphs. Technical report, HP Labs, Palo Alto, CA, USA, 2012.
- [50] John W Ratcliff, David Metzener, et al. Pattern matching: The gestalt approach. *Dr. Dobbs’s Journal*, 13(7):46, 1988.
- [51] I. S. Rozhkov and N. V. Loukachevitch. Prompts in Few-Shot Named Entity Recognition. *Pattern Recognition and Image Analysis*, 33(2):122–131, June 2023. ISSN 1054-6618, 1555-6212. doi: 10.1134/S1054661823020104. URL <https://link.springer.com/10.1134/S1054661823020104>.
- [52] Vivek Kumar Singh, Prashasti Singh, Mousumi Karmakar, Jacqueline Leta, and Philipp Mayr. The Journal Coverage of Web of Science, Scopus and Dimensions: A Comparative Analysis. *Scientometrics*, 126(6):5113–5142, 2021. doi: 10.1007/s11192-021-03948-5. URL <http://arxiv.org/abs/2011.00223>.
- [53] Carl Staelin, Michael Elad, Darryl Greig, Oded Shmueli, and Marie Vans. Biblio: automatic meta-data extraction. *International Journal of Document Analysis and Recognition (IJDAR)*, 10(2):113–126, 2007.
- [54] Endang Suryawati and Dwi H Widyanoro. Combination of heuristic, rule-based and machine learning for bibliography extraction. In *2017 5th International Conference on Instrumentation, Communications, Information Technology, and Biomedical Engineering (ICICI-BME)*, pages 276–281. IEEE, 2017.
- [55] Zhen Tan, Alimohammad Beigi, Song Wang, Ruocheng Guo, Amrita Bhattacharjee,

- Bohan Jiang, Mansooreh Karami, Jundong Li, Lu Cheng, and Huan Liu. Large language models for data annotation: A survey. *arXiv preprint arXiv:2402.13446*, 2024.
- [56] Dominika Tkaczyk, Pawel Szostek, Piotr Jan Dendek, Mateusz Fedoryszak, and Lukasz Bolikowski. Cermine – automatic extraction of metadata and references from scientific literature. In *2014 11th IAPR International Workshop on Document Analysis Systems*, pages 217–221, 2014. doi: 10.1109/DAS.2014.63.
- [57] Dominika Tkaczyk, Andrew Collins, Paraic Sheridan, and Joeran Beel. Evaluation and comparison of open source bibliographic reference parsers: A business use case. *arXiv preprint arXiv:1802.01168*, 2018.
- [58] Dominika Tkaczyk, Andrew Collins, Paraic Sheridan, and Joeran Beel. Machine learning vs. rules and out-of-the-box vs. retrained: An evaluation of open-source bibliographic reference and citation parsers. In *Proceedings of the 18th ACM/IEEE on Joint Conference on Digital Libraries*, JCDL ’18, page 99–108, New York, NY, USA, 2018. Association for Computing Machinery. ISBN 9781450351782. doi: 10.1145/3197026.3197048. URL <https://doi.org/10.1145/3197026.3197048>.
- [59] Dominika Tkaczyk, Rohit Gupta, Riccardo Cinti, and Joeran Beel. Parsrec: A novel meta-learning approach to recommending bibliographic reference parsers. *arXiv preprint arXiv:1811.10369*, 2018.
- [60] Ioannis Tsochantaridis, Thorsten Joachims, Thomas Hofmann, Yasemin Altun, and Yoram Singer. Large margin methods for structured and interdependent output variables. *Journal of machine learning research*, 6(9), 2005.
- [61] Sanh Victor, Webson Albert, Raffel Colin, Bach Stephen, Sutawika Lintang, Alyafeai Zaid, Chaffin Antoine, Stiegler Arnaud, Raja Arun, Dey Manan, et al. Multitask prompted training enables zero-shot task generalization. In *International Conference on Learning Representations*, 2022.
- [62] Shuhe Wang, Xiaofei Sun, Xiaoya Li, Rongbin Ouyang, Fei Wu, Tianwei Zhang, Jiwei Li, and Guoyin Wang. GPT-NER: Named Entity Recognition via Large Language Models, October 2023. URL <http://arxiv.org/abs/2304.10428>. arXiv:2304.10428 [cs].
- [63] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, et al. Emergent abilities of large language models. *arXiv preprint arXiv:2206.07682*, 2022.
- [64] Xiang Wei, Xingyu Cui, Ning Cheng, Xiaobin Wang, Xin Zhang, Shen Huang, Pengjun Xie, Jinan Xu, Yufeng Chen, Meishan Zhang, Yong Jiang, and Wenjuan Han. Zero-Shot Information Extraction via Chatting with ChatGPT, February 2023. URL <http://arxiv.org/abs/2302.10205>. arXiv:2302.10205 [cs].
- [65] Jason Weston, Chris Watkins, et al. Support vector machines for multi-class pattern recognition. In *Esann*, volume 99, pages 219–224, 1999.
- [66] Paul Wouters. Eugene Garfield (1925–2017). *Nature*, 543(7646):492–492, 2017. doi: 10.1038/543492a. URL <https://doi.org/10.1038%2F543492a>.
- [67] Ping Yin, Ming Zhang, ZhiHong Deng, and DongQing Yang. Metadata extraction from bibliographies using bigram hmm. In *International Conference on Asian Digital Libraries*, pages 310–319. Springer, 2004.
- [68] Qing Zhang, Yong-Gang Cao, and Hong Yu. Parsing citations in biomedical articles using conditional random fields. *Computers in biology and medicine*, 41(4):190–194, 2011.
- [69] Xiaoli Zhang, Jie Zou, Daniel X Le, and George R Thoma. A structural svm approach for reference parsing. *BMC bioinformatics*, 12(3):1–7, 2011.
- [70] Wenxuan Zhou, Sheng Zhang, Yu Gu, Muhao Chen, and Hoifung Poon. UniversalNER: Targeted Distillation from Large Language Models for Open Named Entity Recognition,

August 2023. URL <http://arxiv.org/abs/2308.03279>. arXiv:2308.03279 [cs].

- [71] Jie Zou, Daniel Le, and George R Thoma. Locating and parsing bibliographic references in html medical articles. *International Journal on Document Analysis and Recognition (IJDAR)*, 13(2):107–119, 2010.